

Prevalence of ESBL and Carbapenemase-Producing Gram-Negative Bacteria Isolated from Diabetic Foot Infections

Fatma SADDAR^{1*2}, Mourad BOUKOUCHA²

¹Biomolecules and Application Laboratory, Faculty of Exact Sciences and Natural and Life Sciences, Echahid Cheikh Larbi Tebessi University, 12002 Tébessa, Algeria

²Department of Applied Biology, Faculty of Exact Sciences and Natural and Life Sciences, Echahid Cheikh Larbi Tebessi University, 12002 Tébessa, Algeria

Fatma.saddar@univ-tebessa.dz

Abstract

This study aimed to evaluate the prevalence of extended-spectrum β -lactamase (ESBL) and carbapenemase production among Gram-negative bacteria isolated from DFIs.

Sterile swabs were used to collect samples from foot ulcers. Bacterial identification, antimicrobial susceptibility testing, and phenotypic detection of ESBL and carbapenemase production were performed using the VITEK 2 Compact system.

Among 80 tested isolates, 53.9% (43/80) were phenotypically ESBL producers. The highest ESBL frequencies were observed in *Klebsiella* spp. 75% (6/11), *Pseudomonas aeruginosa* 64.3% (16/26), *Escherichia coli* 57.8% (10/18), and *Proteus mirabilis* (25%, 4/16). Additionally, 27.9% (22/80) of isolates were carbapenemase producers, mainly *Klebsiella* spp 62.5% (7/11), *P. aeruginosa* 23.5% (6/26), and *E. coli* 15.8% (3/18).

Background-objectives

Diabetic foot infections (DFIs) represent a major clinical and economic burden for diabetic patientsThey are among the most serious metabolic complications of diabetes and may rapidly progress to irreversible septic gangrene requiring limb amputation (**Datta *et al*, 2019**) Optimizing antibiotic therapy for DFIs relies on understanding the antibiotic susceptibility patterns of the involved bacterial pathogens(**Saleem *et al*, 2025**).

This study aimed to evaluate the prevalence of extended-spectrum β -lactamase (ESBL) and carbapenemase production among Gram-negative bacteria isolated from DFIs

1-Methods

1-Bacterial strains

Isolation of bacterial strains Sterile swabs were used to collect samples from foot ulcers



Fig1:infected diabetic foot

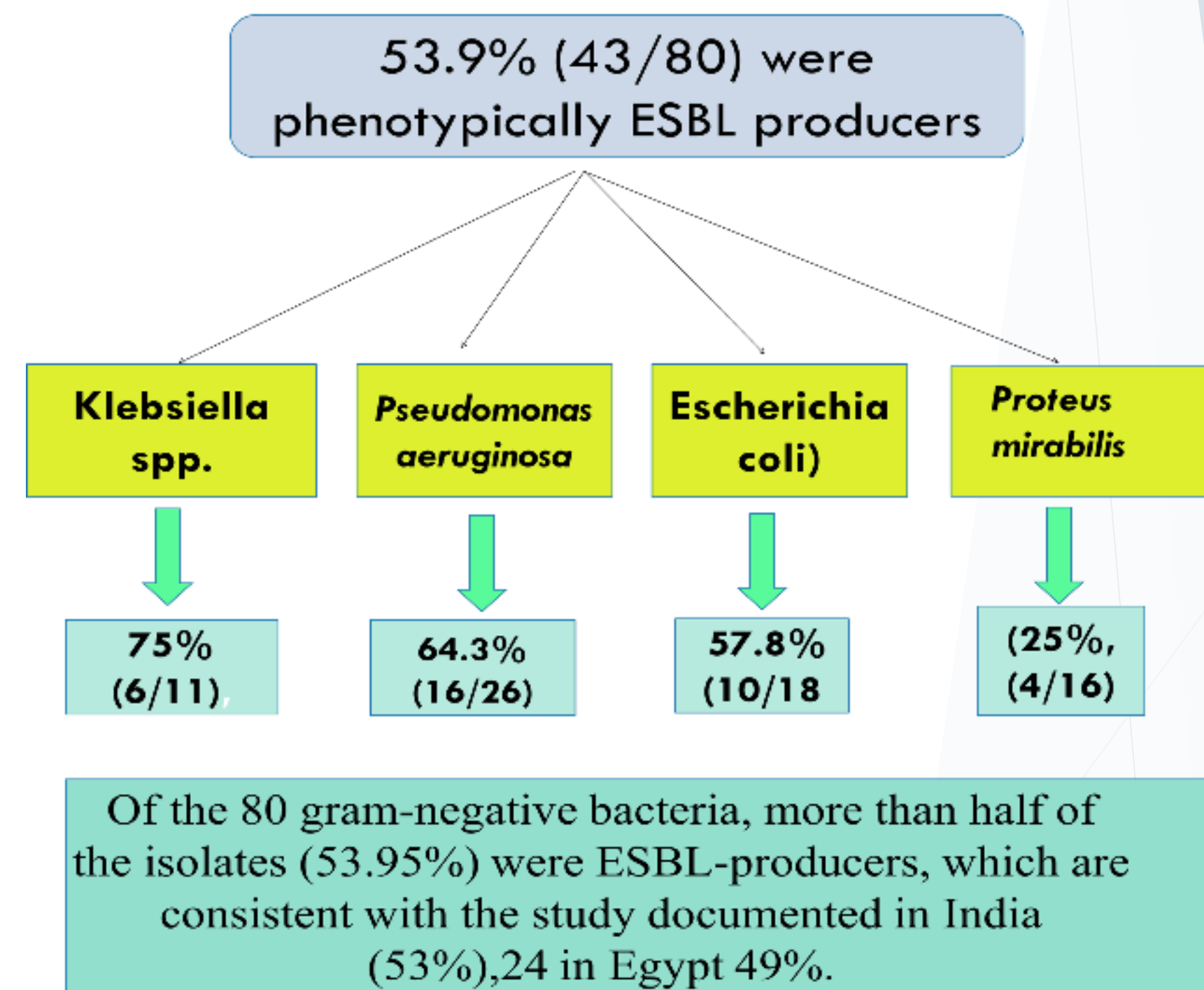
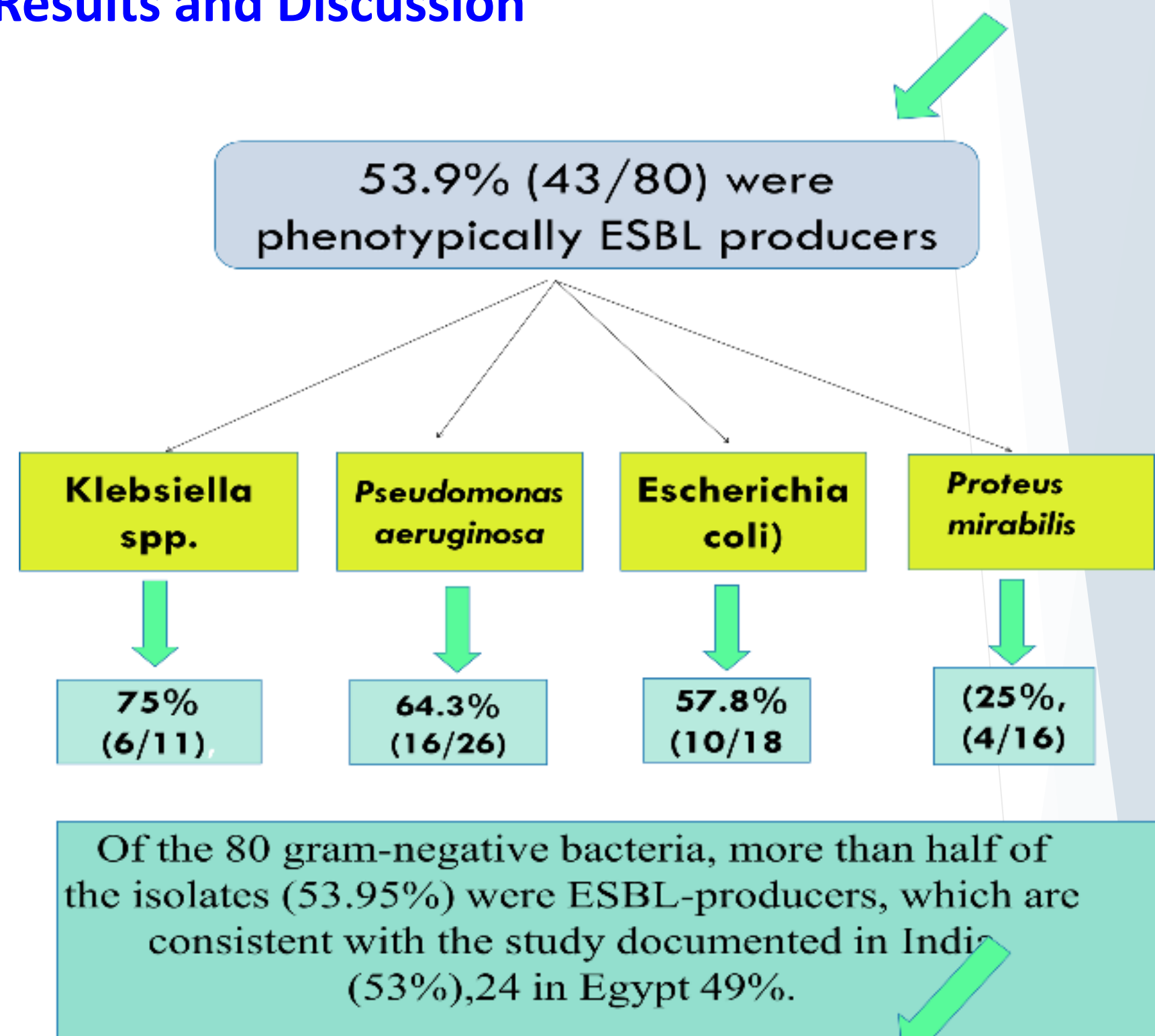
2/Bacterial identification, antimicrobial susceptibility testing, and phenotypic detection of ESBL and carbapenemase production

Bacterial identification, antimicrobial susceptibility testing, and phenotypic detection of ESBL and carbapenemase production were performed using the VITEK 2 Compact system.



Fig2: identification cards vitek 2

2-Results and Discussion



Conclusion

These findings highlight the importance of microbiological testing and antimicrobial susceptibility profiling in the management of diabetic foot infections, especially given the high prevalence of multidrug-resistant ESBL- and carbapenemase-producing microorganisms

Bibliographic references

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